

Interpretable, Inclusive, and Immersive Interaction for Ubiquitous AI-infused Physical Systems

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Abstract

As ubiquitous computers permeate our daily lives, we increasingly encounter AI-infused physical systems, where physical entities are augmented with sensing, understanding, and actuating capabilities. While they have the potential to impact our physical reality, their proactive nature also poses unique challenges for users. Interpreting the reasoning behind the systems' behavior can be difficult due to the interplay between computations and physical exertion. Ensuring inclusive interactions is also a key issue, as the interactions and interfaces may not inherently accommodate users with diverse abilities, backgrounds, or preferences. Achieving immersive interactions that blend digital and physical contexts requires understanding human behavior in dynamic environments. To address these interconnected challenges, we propose a workshop for interdisciplinary collaboration among researchers and practitioners from ubiquitous computing, HCI, AI, robotics, design, and accessibility domains. This workshop explores novel approaches, methods, design guides, and case studies around these issues.

CCS Concepts

• Human-centered computing → Ubiquitous computing.

Keywords

Ubiquitous computing; Human-AI Interaction; AI-infused Systems

ACM Reference Format:

Gwangbin Kim, Minwoo Seong, Dohyeon Yeo, Yumin Kang, and SeungJun Kim. 2024. Interpretable, Inclusive, and Immersive Interaction for Ubiquitous AI-infused Physical Systems. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3675094.3677569>

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UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia
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ACM ISBN 979-8-4007-1058-2/24/10
<https://doi.org/10.1145/3675094.3677569>

1 Background

The recent advancements in AI have led to the automation of an increasing number of human tasks. The fields of ubiquitous computing and interaction design are accelerating the integration of AI into our daily lives by incorporating AI into everyday systems that involve human interaction. To foster safe, reliable, and comfortable human experiences with these AI systems, discussions regarding human-AI interaction have gained significant attention.

The infusion of AI extends beyond digitized data processing; AI is integrated with physical sensing and actuating capabilities, ranging from routine goods to specialized AI-infused products. Despite the comfort they provide, they also pose interaction challenges. First, complex AI models make interactions uninterpretable and may exclude certain groups with the lack of data from the elderly [23] and with the limited usability for people with disabilities [22] (the challenge of interpretable and inclusive interaction). Second, physical system constraints limit integrated AI capabilities, while extended reality offers solutions for immersive experiences with augmented sensory experiences [7, 11]. Yet, designing immersive interactions poses challenges, such as ensuring seamless integration, user comfort, and safety (the challenge of immersive interaction).

In response, discussions must focus on interactions with AI-infused physical systems, ensuring interpretability and accessibility for inclusive experiences. The ubiquity and impact of these systems underscore the importance of this focus. An interdisciplinary approach is needed to explore strategies for creating interpretable, inclusive, and immersive interactions, enhancing trust and user experience while enabling AI to transcend physical limitations.

2 Designing Interpretable, Inclusive, and Immersive Interactions

AI increasingly impact physical reality through ubiquitous sensing and actuation, tracking human position [28], actions [27], and states [5, 13]. They assist our lives in various ways, from minor comforts like alleviating dry eyes [2], to vital tools for living such as low vision assistance [10], and wayfinding design [15]. While they benefit humans by expanding our capabilities, they also pose potential risks to our experience. Given the growing roles and complexity, ensuring interpretable and inclusive interactions is crucial.

As the roles and complexity of AI-infused physical systems expand, so do their interaction spaces. While most applications have traditionally focused on indoor scenarios suitable for computing, interaction spaces are now ubiquitous, such as vehicles on roads [12, 14], and becoming increasingly mobile due to wearable technologies [8]. Moreover, the interaction spaces for AI-infused physical systems are also expanding in terms of reality. Thus, for interpretable and inclusive co-existence with AI-powered physical entities across physical and virtual realities, their design under immersive interaction environments should be discussed.

2.1 Objectives

We propose the following questions to guide the workshop theme and set long-term agendas:

- **RQ1:** How can we design interpretable interactions for AI-infused systems that bridge the gap between human expectation and actual system behavior for user trust and adoption?
- **RQ2:** What challenges and opportunities arise in creating inclusive and accessible interactions with AI-infused systems to accommodate diverse needs, preferences, and abilities while promoting long-term user well-being?
- **RQ3:** How can we leverage emerging technologies and extended reality support to enable natural and intuitive interaction with AI-infused physical systems?
- **RQ4:** What are the ethical, social, and cultural implications of AI-infused systems, and how can we develop practices and principles that align with human values?

Research Contributions. To answer the research questions, we invite contributions including but not limited to:

- Artifacts and prototypes showcasing interaction with networked or embedded intelligence in ‘physical’ systems
- Positions exploring novel interaction paradigms, design principles, and frameworks for AI-infused ‘physical’ systems
- Case studies assessing the interaction quality and user experience of AI-infused ‘physical’ systems
- Enabling technologies, platforms, and infrastructures supporting the development of AI-infused ‘physical’ systems
- User studies investigating user needs, preferences, and characteristics in interacting with AI-infused ‘physical’ systems

Algorithmic Contributions. We provide an algorithm track to leverage open datasets for researchers with limited resources. While open to all types of existing datasets, we can provide guidance during the workshop for those released by the organizers (**bold**). All tracks will have the same review process. Example datasets:

- **Engagnition: A multi-dimensional dataset for engagement recognition of children with autism spectrum disorder [17]**
- **MultiSenseBadminton: Wearable Sensor-Based Biomechanical Dataset for Evaluation of Badminton Performance [25]**

3 Organizers

3.1 Organizing Committee

- **Gwangbin Kim** researches intelligible interaction for ubiquitous human-AI systems in contexts such as automated

vehicles. He holds a B.S. in Mechanical Engineering and an M.S. in Robotics from GIST, where he is pursuing his Ph.D. Previously, he worked as a research engineer at SOSLAB.

- **Minwoo Seong** researches human motion reconstruction. He wrote Scientific Data papers on multimodal physical activities. Seong holds a B.S. in Mech. Engineering and an M.S. in Robotics from GIST, where he is pursuing his Ph.D. He received awards from KCC, CHI, and IJCAI Challenge.
- **Dohyeon Yeo** researches immersive experiences with interactive systems. He holds a B.S. in control and automation engineering from Korea Maritime and Ocean University and an M.S. in Robotics from GIST, where he is pursuing his Ph.D. He received paper awards from KHCI, CHI, and UbiComp.
- **Yumin Kang** designs interactive systems to aid user decision making. She earned her B.S. in Media Tech. from Hanyang University ERICA and M.S. in Robotics from GIST, where she is pursuing her Ph.D. Her research on pedestrian behavior with automated vehicles received an AutoUI paper award.
- **SeungJun Kim** is an associate professor at GIST, directing the human-centered intelligent systems lab. Before that, he was a special faculty at the HCI Institute, Carnegie Mellon University, and led research projects on XR, robotics, and AI. Kim holds a B.S. in Electrical Engineering from KAIST and an M.S. and Ph.D. in Mechatronics from GIST. His research focuses on sensory intelligence and augmentation through multimodal XR and was recognized with paper awards from CHI, UbiComp, ISMAR, and AutoUI.

3.2 Steering Committee

- **Ahmed Elsharkawy** is an assistant professor of electronic engineering at Menoufia University and a post-doc at GIST. He received a B.S. degree in Industrial Electronics and Control Engineering from Menoufia University, an M.S. in Mechatronics and a Ph.D. in Robotics from GIST. His research interests include HRI, haptics, and multimodal XR.
- **Euseok Hwang** is a professor of EECS at GIST. He holds B.S. and M.S. from Seoul National University and M.S. and Ph.D. from Carnegie Mellon University. Before GIST, he worked at Daewoo Electronics and at LSI (now Broadcom). He was a visiting scholar at the University of Michigan in 2021-22. His interests are signal processing, IoT, and smart grids.
- **Jin-Hyuk Hong** is an associate professor at the School of Integrated Technology, GIST. He received a B.S., M.S., and Ph.D. from Yonsei University. His research interests include context awareness, machine learning, and interaction design. Before joining GIST, he was a special faculty at the HCI Institute, Carnegie Mellon University and worked as a Principal Researcher at Naver and Samsung Electronics.
- **Kyung-Joong Kim** is an associate professor and the director of the Korea Culture Technology Research Institute at GIST. He received B.S., M.S., and Ph.D. degrees in computer science from Yonsei University. Previously, he was a postdoc at Cornell University and an associate professor at Sejong University. He conducts research on AI, games, and robotics. Kim has served as general chair of the IEEE CoG and associate editor of IEEE Transactions on Games.

- **Jieun Lee** is a post-doc at GIST. She holds a B.S. in biomedical engineering from Eulji University, an M.S. in Robotics from DGIST, and a Ph.D. in Robotics from GIST with presidential citation for outstanding research. Her research interests include multimodal XR interfaces and redirected walking.
- **Yiyue Luo** is an assistant professor of ECE at the university of Washington. She earned her Ph.D. in EECS from MIT, advised by Prof. Wojciech Matusik and Prof. Tomas Palacios. Her research lies at the intersection of digital fabrication, HCI, and AI and has been published in interdisciplinary journals, and top HCI, robotics, and AI venues. She received fellowships from MathWorks, Google, and Accenture, and was listed as Forbes 30 under 30 North American 2024.
- **Byung-Cheol Min** is an Associate Professor of Computer and Information Technology at Purdue University, directing the SMART Lab. His HRI research has been recognized with the NSF CAREER Award and the Purdue University Faculty Scholar. He received his Ph.D. from Purdue University and an M.S. and B.S. from Kyung Hee University. Before joining Purdue, he was a postdoc at the CMU Robotics Institute.

4 Workshop Plans

Workshop Structure and Format. The workshop will be full-day (Table 1), with two invited talks, one by a senior researcher with a visionary outlook, and the other for another senior keynote or by junior researchers to cater to diverse voices in the field. The workshop’s main program will feature paper presentations.

Pre-workshop preparation. We will form a program committee consisting of experts from academia and industry. Upon acceptance after review by PC members, we will organize the workshop schedule and work with attendees to accommodate their needs and explore potential accessibility solutions within available resources.

Review and Publication Process. Each submission will undergo a peer review with at least two reviews. As this workshop is intended to stimulate discussions on future research agendas, the review criteria will emphasize workshop relevance, novelty, and rigorosity. All accepted papers will be published in the ACM DL.

Justice, Equity, Diversity, and Inclusion (JEDI) Statement. Our workshop is committed to fostering a welcoming and inclusive environment for all attendees. We will actively encourage and facilitate participation from underrepresented groups by promoting the

workshop through ACM chapters in diverse groups and countries. We will provide guidelines for accessible materials to participants and work closely with the conference chairs to create an inclusive experience for all attendees, considering venue accessibility, room layouts, seating arrangements, etc.

5 Editorial from the Program Committee

As AIs permeate our physical world, their impact on our lives grows significantly. The ubiquity and complexity of AI-infused physical systems call for interpretable, inclusive, and immersive interaction. In this workshop, 45 authors from 12 institutions explore approaches to achieve these goals across diverse application areas.

Some papers focus on making AI systems’ actions transparent and understandable, introducing additional modalities. Hwang et al. [6] develop a biomechanical approach for force feedback in VR teleoperation, allowing users to understand the movement of remote robot arms through augmented somatosensations.

Conversely, several papers aid AI-infused systems in understanding human states. Seong et al. [24] explore tactile sensing to infer human 3D posture. Elsharkawy et al. [4] investigate how computers inducing specific human postures can reduce motion sickness during in-car VR entertainment. Moon et al. [18] propose improved self-annotation in affective computing, more accurately capturing nuanced changes in human emotional states during gameplay.

On the inclusivity front, multiple works focus on accessible AI interactions for diverse user groups. Choi et al. [1] explore generative AIs to support song-signing creation for deaf users. Noh and Hong [20] present a conversational system to provide reminiscence therapy for people with dementia. Kim et al. [16] suggest application areas of engagement recognition systems for autistic children for proactive intervention in effective and efficient edutainment.

Just as AI-infused systems are designed to support people with accessibility needs, they can also enhance the experience of caregivers. Okuda et al. [21] propose an activity prediction algorithm for improved caring efficiency for dependent individuals. Similarly, Edelman et al. [3] explore the use of AI for MRI training to ensure the stationary posture of patients, improving imaging quality for prescriptions given the high cost of MRI diagnosis. These healthcare-focused papers demonstrate how AI implementation can enhance medical care and accessibility for diverse patient populations.

Several papers address immersive interaction. Um et al. [26] induce walking sensations in seated VR users through electrical muscle stimulation. Janaka et al. [9] provide an open-source platform for wearable assistants for body-proximal interactions, with three preliminary use case studies. Mosur et al. [19] explore optimal AR display while walking and reading for wearable, everyday use.

These 12 papers represent AI-infused physical systems in healthcare, entertainment, and workplace settings, highlighting the potential and challenges of AI systems that integrate into users’ lives.

Acknowledgments

This work was in part supported by the GIST-MIT Research Collaboration grant funded by the GIST in 2024. This work was in part supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (MSIT) (RS-2024-00343397).

Table 1: Proposed Workshop Schedule

Time	Activity
09:00-09:15	Opening: Welcome and introduction of the organizers, participants, and schedule
09:15-10:15	Invited Talk #1: Keynote by an invited expert or three invited talks of workshop-relevant full papers
10:15-10:45	Coffee Break
10:45-11:45	Paper Presentations #1: 6 paper presentations
11:45-13:30	Networking and Lunch Break
13:30-14:30	Invited Talk #2: Keynote by an invited expert or three invited talks of workshop-relevant full papers
14:30-15:00	Coffee Break
15:00-16:00	Paper Presentations #2: 6 paper presentations
16:00-16:30	Closing: Wrap up and awards

References

- [1] YouJin Choi, JaeYoung Moon, Kyung-Joong Kim, and Jin-Hyuk Hong. 2024. Exploring the Potential of Generative AI in Song-Signing. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3678378>
- [2] Artem Dementyev and Christian Holz. 2017. DualBlink: A Wearable Device to Continuously Detect, Track, and Actuate Blinking For Alleviating Dry Eyes and Computer Vision Syndrome. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 1, 1, Article 1 (mar 2017), 19 pages. <https://doi.org/10.1145/3053330>
- [3] Valentina Edelman, Irina Rabaev, and Hadas Chassidim. 2024. Reducing Patient Movement in MRI: A Pilot Study of a Training System. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3678385>
- [4] Ahmed Elsharkawy, Aya Ataya, Dohyeon Yeo, Minwoo Seong, Seokhyun Hwang, Joseph DelPreto, Wojciech Matusik, Daniela Rus, and SeungJun Kim. 2024. Adaptive In-Vehicle Virtual Reality for Reducing Motion Sickness: Manipulating Passenger Posture During Driving Events. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3678381>
- [5] Eija Haapalainen, SeungJun Kim, Jodi F. Forlizzi, and Anind K. Dey. 2010. Psychophysiological measures for assessing cognitive load. In *Proceedings of the 12th ACM International Conference on Ubiquitous Computing* (Copenhagen, Denmark) (*UbiComp '10*). Association for Computing Machinery, New York, NY, USA, 301–310. <https://doi.org/10.1145/1864349.1864395>
- [6] Seokhyun Hwang, Seungjun Kang, Jeongseok Oh, Jeongju Park, Semoo Shin, Yiyue Luo, Joseph DelPreto, Wojciech Matusik, Daniela Rus, and SeungJun Kim. 2024. Proposal of a Framework for Enhancing Teleoperation Experience with Biomechanical Simulation-Based Electrical Muscle Stimulation in Virtual Reality. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3675094.3678380>
- [7] Seokhyun Hwang, Jeongseok Oh, Seungjun Kang, Minwoo Seong, Ahmed Ibrahim Ahmed Mohamed Elsharkawy, and SeungJun Kim. 2024. ErgoPulse: Electrifying Your Lower Body With Biomechanical Simulation-based Electrical Muscle Stimulation Haptic System in Virtual Reality. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, New York, NY, USA, Article 416, 21 pages. <https://doi.org/10.1145/3613904.3642008>
- [8] Abhinandan Jain, Adam Haar Horowitz, Felix Schoeller, Sang-won Leigh, Pattie Maes, and Misha Sra. 2020. Designing Interactions Beyond Conscious Control: A New Model for Wearable Interfaces. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 4, 3, Article 108 (sep 2020), 23 pages. <https://doi.org/10.1145/3411829>
- [9] Nuwan Janaka, Shengdong Zhao, David Hsu, Sherisse Tan Jing Wen, and Koh Chun Keat. 2024. TOM: A Development Platform For Wearable Intelligent Assistants. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3675094.3678382>
- [10] Taewoo Jo, Dohyeon Yeo, Gwangbin Kim, Seokhyun Hwang, and SeungJun Kim. 2024. WatchCap: Improving Scanning Efficiency in People with Low Vision through Compensatory Head Movement Stimulation. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 8, 2, Article 47 (may 2024), 32 pages. <https://doi.org/10.1145/3411825>
- [11] Seungjun Kang, Gwangbin Kim, and SeungJun Kim. 2023. Giant Finger: A Novel Visuo-Somatosensory Approach to Simulating Lower Body Movements in Virtual Reality. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 233–242. <https://doi.org/10.1109/ISMAR59233.2023.00038>
- [12] Gwangbin Kim, Seokhyun Hwang, Minwoo Seong, Dohyeon Yeo, Daniela Rus, and SeungJun Kim. 2024. TimelyTale: A Multimodal Dataset Approach to Assessing Passengers' Explanation Demands in Highly Automated Vehicles. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 8, 3, Article 109 (sep 2024), 60 pages. <https://doi.org/10.1145/3678544>
- [13] Gwangbin Kim, Jieun Lee, Dohyeon Yeo, Eunsol An, and SeungJun Kim. 2023. Physiological Indices to Predict Driver Situation Awareness in VR. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing* (, Cancun, Quintana Roo, Mexico.) (*UbiComp/ISWC '23 Adjunct*). Association for Computing Machinery, New York, NY, USA, 40–45. <https://doi.org/10.1145/3594739.3610687>
- [14] Gwangbin Kim, Dohyeon Yeo, Taewoo Jo, Daniela Rus, and SeungJun Kim. 2023. What and When to Explain? On-road Evaluation of Explanations in Highly Automated Vehicles. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7, 3, Article 104 (sep 2023), 26 pages. <https://doi.org/10.1145/3610886>
- [15] Gwangbin Kim, Dohyeon Yeo, Jieun Lee, and SeungJun Kim. 2023. Simulating Urban Element Design with Pedestrian Attention: Visual Saliency as Aid for More Visible Wayfinding Design. *Land* 12, 2 (2023). <https://doi.org/10.3390/land12020394>
- [16] Won Kim, Minwoo Seong, Joseph DelPreto, Wojciech Matusik, Daniela Rus, and SeungJun Kim. 2024. Exploring Potential Application Areas of Artificial Intelligence-Infused System for Engagement Recognition: Insights from Special Education Experts. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3675094.3678376>
- [17] Won Kim, Minwoo Seong, Kyung-Joong Kim, and SeungJun Kim. 2024. Engagnition: A multi-dimensional dataset for engagement recognition of children with autism spectrum disorder. *Scientific Data* 11, 1 (2024), 299.
- [18] JaeYoung Moon, YouJin Choi, Georgios N. Yannakakis, and Kyung-Joong Kim. 2024. Progress of PREFAB: Preference based Self-Annotation on a Low-Budget. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3678379>
- [19] Priyanka Mosur, Ethan Kimmel, Parth Arora, Rajandeep Singh, Atharva Rajesh Madiwale, Jenna Kang, and Thad Starner. 2024. Stepping into AR: Exploring Optimal Positioning for Monocular Head-Worn Displays for Reading on the Go. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3675094.3678383>
- [20] Yeo-Gyeong Noh and Jin-Hyuk Hong. 2024. Know Me Inside-Out: Conversational and Quiz-Based System for Reminiscence Therapy for People with Dementia. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3675094.3678377>
- [21] Ryuichiro Okuda, Qingxin Xia, Takuya Maekawa, Takahiro Hara, and Sozo Inoue. 2024. Preliminary Investigation of Activity Prediction in Nursing Homes using Activity History with Erroneous Time Stamps. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3675094.3678384>
- [22] Ozlem Ozmen Garibay, Brent Winslow, Salvatore Andolina, Margherita Antona, Anja Bodenschatz, Constantinos Coursaris, Gregory Falco, Stephen M Fiore, Ivan Garibay, Keri Grieman, et al. 2023. Six human-centered artificial intelligence grand challenges. *International Journal of Human-Computer Interaction* 39, 3 (2023), 391–437.
- [23] Eunjin Seong and SeungJun Kim. 2020. Designing a Crowdsourcing System for the Elderly: A Gamified Approach to Speech Collection. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (, Honolulu, HI, USA.) (*CHI EA '20*). Association for Computing Machinery, New York, NY, USA, 1–9. <https://doi.org/10.1145/3334480.3382999>
- [24] Minwoo Seong, Gwangbin Kim, Jaehee Lee, Joseph DelPreto, Wojciech Matusik, Daniela Rus, and SeungJun Kim. 2024. Intelligent Seat: Tactile Signal-Based 3D Sitting Pose Inference. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3675094.3678374>
- [25] Minwoo Seong, Gwangbin Kim, Dohyeon Yeo, Yumin Kang, Heesan Yang, Joseph DelPreto, Wojciech Matusik, Daniela Rus, and SeungJun Kim. 2024. MultiSense-Badminton: Wearable Sensor-Based Biomechanical Dataset for Evaluation of Badminton Performance. *Scientific Data* 11, 1 (2024), 343.
- [26] Juwon Um, Eunki Jeon, Yumin Kang, Seungjun Kang, Ahmed Elsharkawy, Joseph DelPreto, Wojciech Matusik, Daniela Rus, and SeungJun Kim. 2024. LegSense: Inducing Walking Sensation in Seated VR by Providing Movement Illusion via Electrical Muscle Stimulation. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Melbourne, VIC, Australia) (*UbiComp Companion '24*). Association for Computing Machinery, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3675094.3678375>
- [27] Fei Wang, Yizhe Lv, Mengdie Zhu, Han Ding, and Jinsong Han. 2024. XRF55: A Radio Frequency Dataset for Human Indoor Action Analysis. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 8, 1, Article 21 (mar 2024), 34 pages. <https://doi.org/10.1145/3643543>
- [28] Binbin Xie, Minhao Cui, Deepak Ganesan, and Jie Xiong. 2024. Wall Matters: Rethinking the Effect of Wall for Wireless Sensing. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7, 4, Article 190 (jan 2024), 22 pages. <https://doi.org/10.1145/3631417>